

 **FACULTY OF ENGINEERING AND MATERIALS SCIENCE**
MECHATRONICS ENGINEERING PROGRAM

MCTR903, Winter Term 10-11

Lecture 1 – Monday, September 27, 2010

Introductory Concepts

Some slides of this lecture are based on Chapter 1 of W. Bolton. *Mechatronics: A Multidisciplinary Approach*. 4th Edition, Pearson, 2008.

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Objectives

When you have finished this lecture you should be able to:

- Trace the origin of mechatronics;
- Understand the key elements of mechatronics systems;
- Relate with everyday examples of mechatronics systems;
- Appreciate how mechatronics integrates knowledge from different disciplines in order to realize engineering and consumer products that are useful in everyday life;
- Recognize the need for models of systems in order to predict their behavior ;
- Describe the various forms and elements of control systems.

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Objectives



Becoming a Mechatronics Engineer

Outline

- What is mechatronics?
- The design process
- Systems
- Measurement systems
- Control systems
- Example of mechatronic systems
- Summary

What is mechatronics?

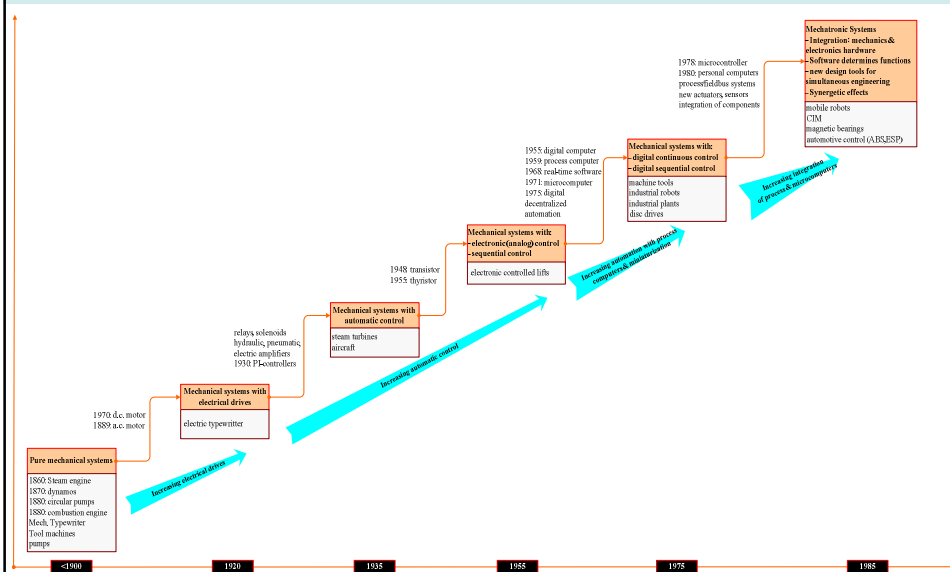


Mechatronic Systems

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What is mechatronics?



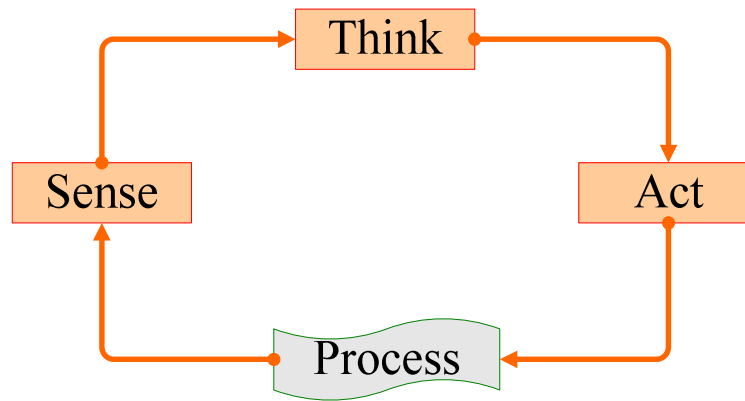
Ref.: R. Isermann, "Mechatronic Design Approach", ch2, in R. Bishop, Mechatronics: An Introduction. CRD Press, 2006.

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What is mechatronics?

• Sense-Think-Act Loop

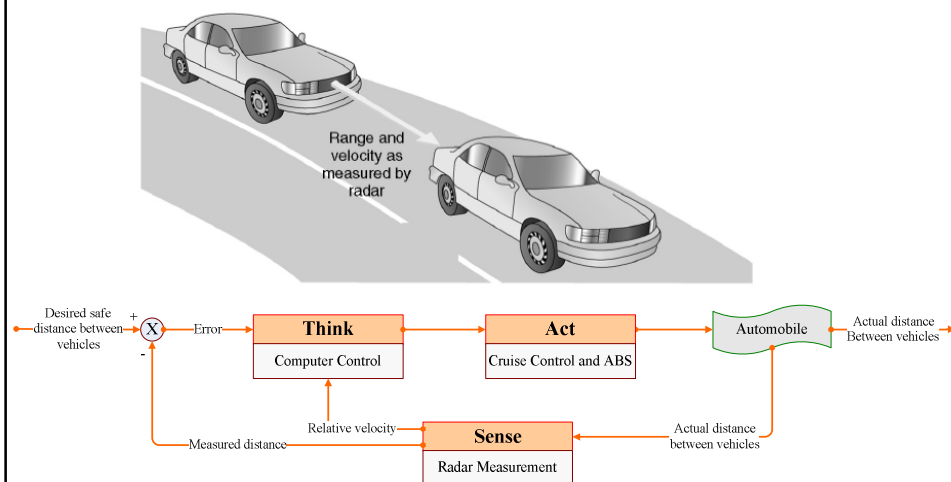


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What is mechatronics?

• Sense-Think-Act Loop

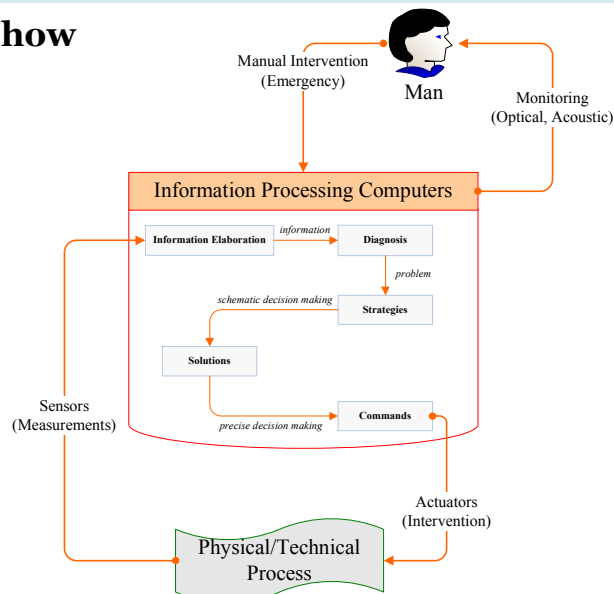


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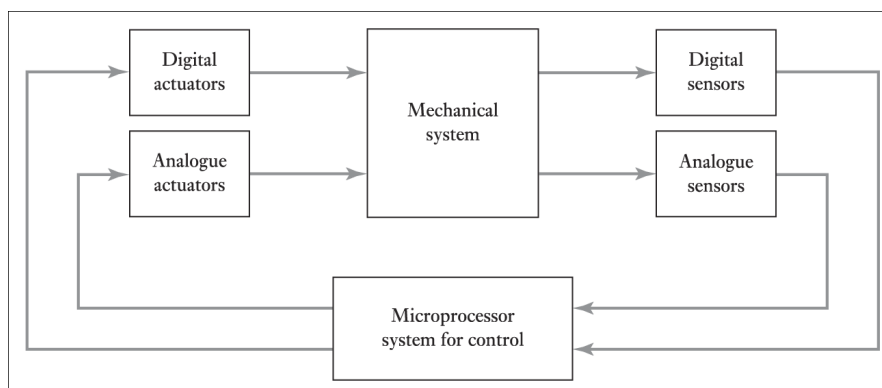
What is mechatronics?

- **Know-how**



What is mechatronics?

- **Mechatronic System**

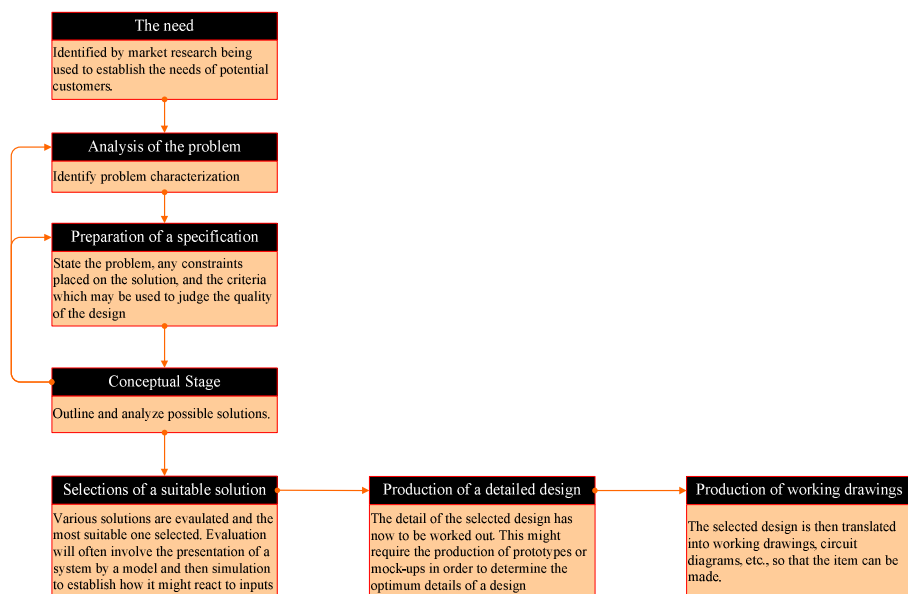


Basic elements of a mechatronic system

Outline

- What is mechatronics?
- **The design process**
- Systems
- Measurement systems
- Control systems
- Example of mechatronic systems
- Summary

Design Process



Outline

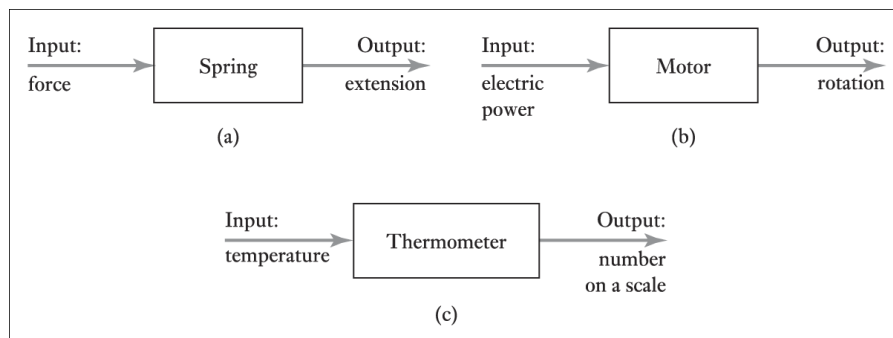
- What is mechatronics?
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Systems

A **system** can be thought of as a box or block diagram which has an input and an output and where we are concerned not with what goes on inside the box but with only the relationship between the output and the input.



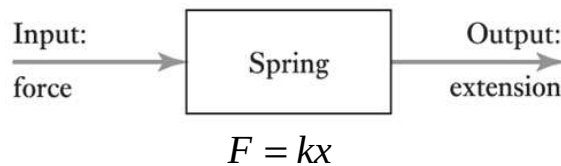
Examples of systems: (a) spring, (b) motor, (c) thermometer

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Systems

The term **modeling** is used when we represent the behavior of a real system by mathematical equations, such equations representing the relationship between the inputs and the outputs from the system



where

F is the input force;

x is the output extension and

k is a constant

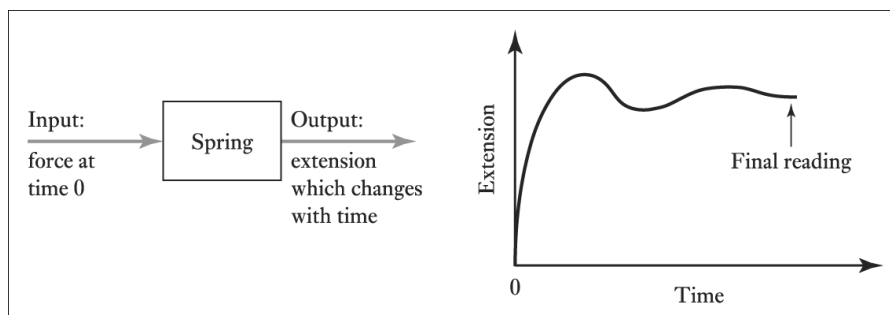
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Systems

The relationship $F = kx$ only describes the relationship between the input and the output when steady-state conditions occur.

When the force is applied it is likely that oscillations will occur before the spring settles down to its steady-state extension value.



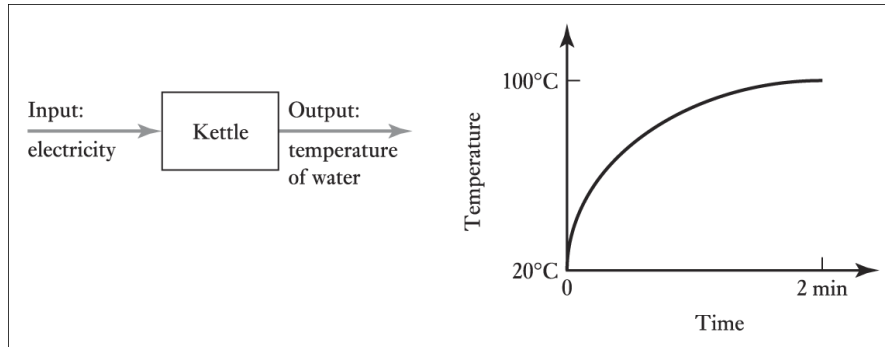
The response to an input for a spring

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Systems

In order to know how systems behave when there are inputs to them, we need to devise models for systems which relate the output to the input so that we can work out, for a given input, how the output will vary with time and what it will settle down to.



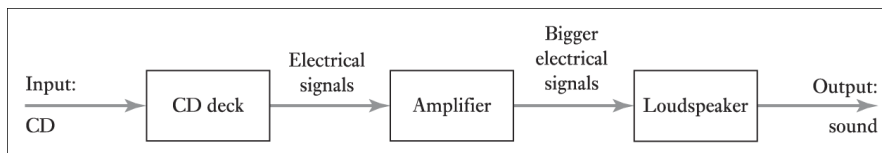
The response to an input for a kettle system

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Systems

Connected system is a system that contains a series of interconnected block, each such block having a specific function.



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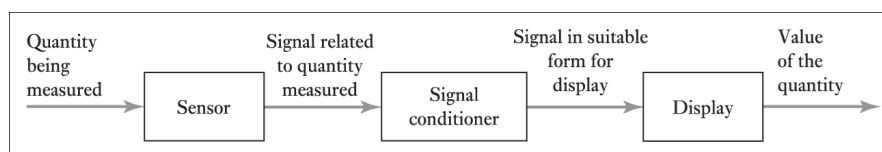
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Outline

- What is mechatronics?
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- **Measurement systems**
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Measurement Systems

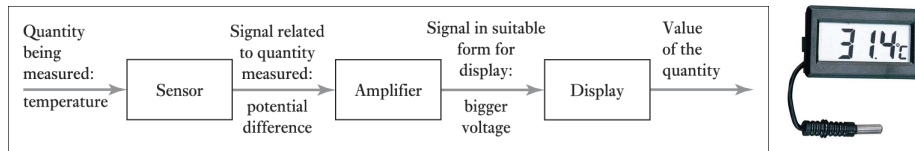
Measurement system can, in general, be considered to be made up of three basic elements:



- **A sensor:** which responds to the quantity being measured by giving as its output a signal which is related to the quantity.
- **A signal conditioner:** takes the signal from the sensor and manipulates it into a condition which is suitable either for display, or, in the case of a control system, for use to exercise control
- **A display:** where the output from the signal conditioner is displayed.

Measurement Systems

Example:



A digital thermometer system

A digital thermometer has an input of temperature to a sensor, probably a semiconductor diode. The potential difference across the sensor is, at constant current, a measure of the temperature. This potential difference is then amplified by an operational amplifier to give a voltage which can be directly drive a display.

Outline

- What is mechatronics?
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Control Systems

A **control system** can be thought of as a system which can be used to:

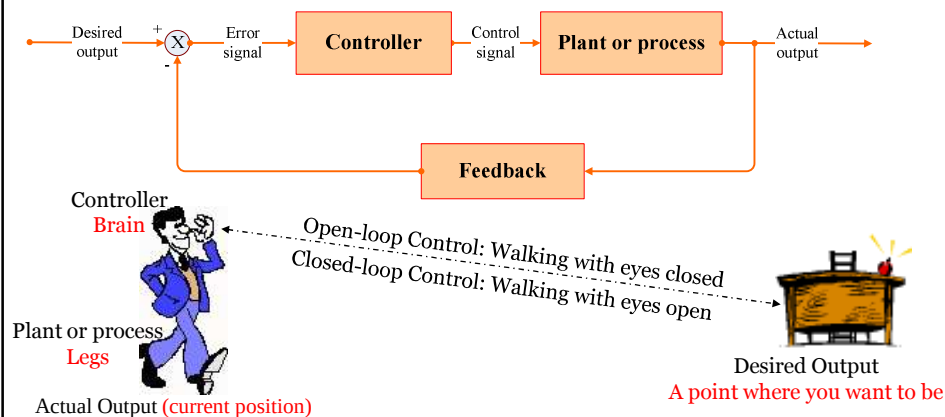
1. Control some variable to some particular value, e.g. a central heating system where the temperature is controlled to a particular value.
2. Control the sequence of events, e.g. washing machine and elevator.
3. Control whether an event occurs or not, e.g. a safety lock on a machine where it cannot be operated until a guard is in position.

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Control Systems

A control system is a combination of components (electrical, mechanical, thermal, or hydraulic) that act together to maintain actual system performance close to a desired set of performance specifications. *Columbia Electronic Encyclopedia*

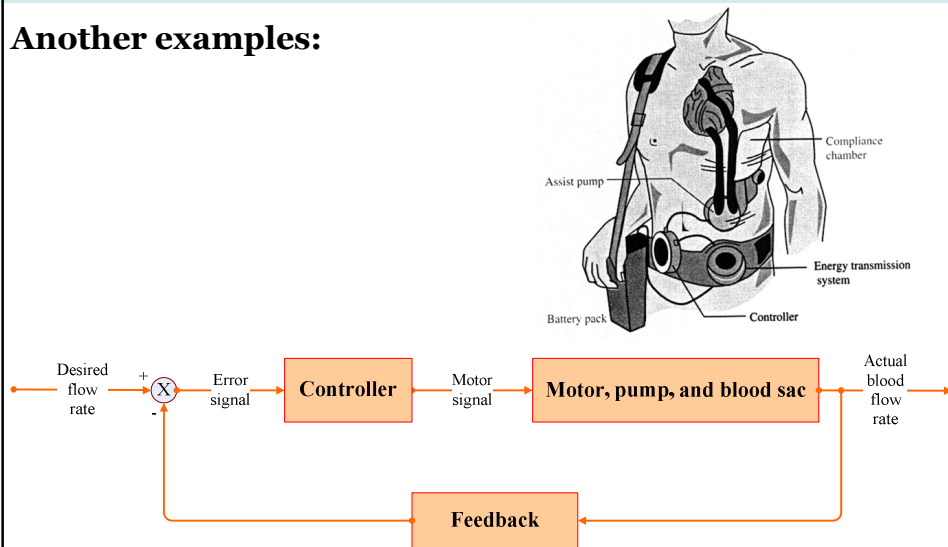


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Control Systems

Another examples:



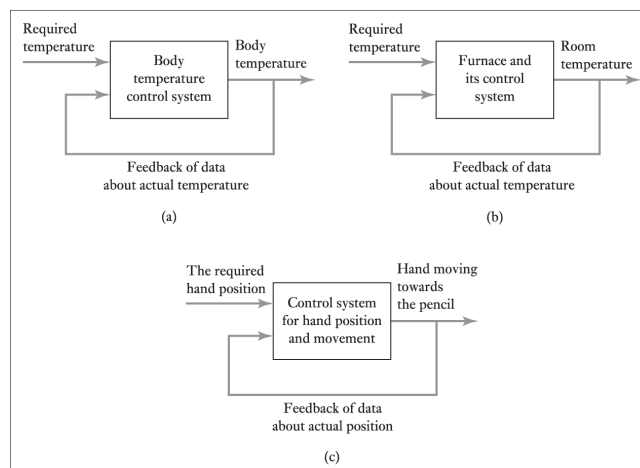
From: R. Dorf and R. Bishop. *Modern Control Systems*. Addison Wesley, 7th edition, 1995, page 706.

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Control Systems

Another examples:



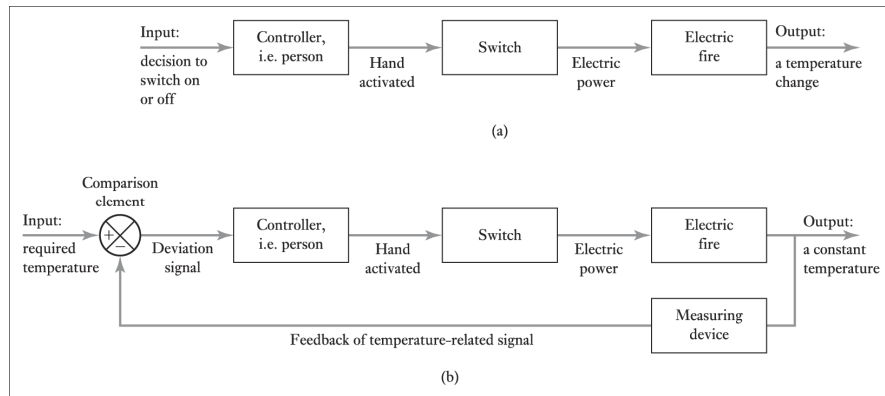
Feedback control: (a) human body temperature, (b) room temperature with central heating, (c) picking up a pencil

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Control Systems

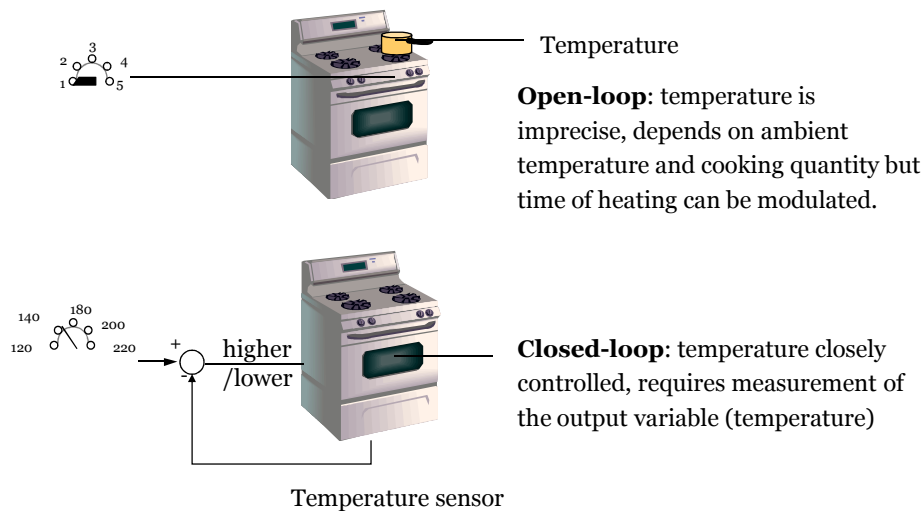
• Open- and closed-loop systems:



Heating a room: (a) an open-loop system, (b) a closed-loop system

Control Systems

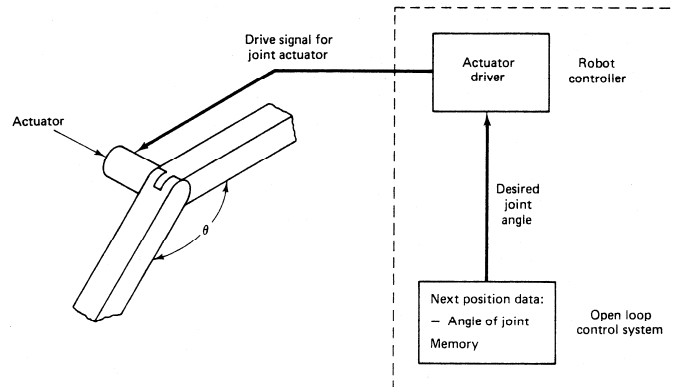
• Open- and closed-loop systems:



Control Systems

• Open- and closed-loop systems:

- Robotic arms as a case study



Open-Loop / Nonservo / stop-to-stop / pick-and-place System

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Control Systems

• Open- and closed-loop systems:

- Robotic arms as a case study

Robots classified as **nonservo or open-loop** do not have position and rate-of-change sensors on any axis; therefore, the controller does not know the position of the arm and tool while the robot is moving from one point to another.

On every axis, however, there is a fixed stop or limit at each extreme of travel that provides the positioning accuracy at that point.

Advantages: the simplicity of the system and the resulting reliability.

Disadvantages: limited and fixed-stop locations and its inability to handle complex manufacturing tasks.

Applications: machine loading-and-unloading.

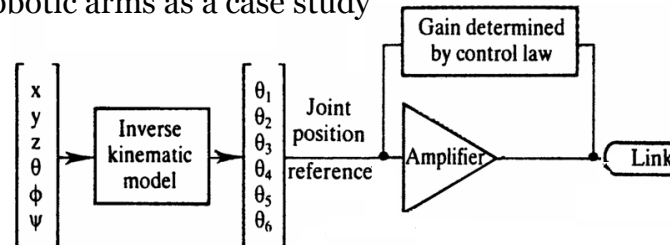
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Control Systems

• Open- and closed-loop systems:

- Robotic arms as a case study



In Cartesian space, we use kinematic model to calculate the joint angles that will place the gripper at the required location in space. The joints are moved to these angles and the robot is assumed to be in the correct Cartesian location, as, generally there is no way of measuring whether it is or not. Open-loop control is **as accurate as the model of the process** (in this case the inverse kinematic model), provided there are no disturbance to cause errors.

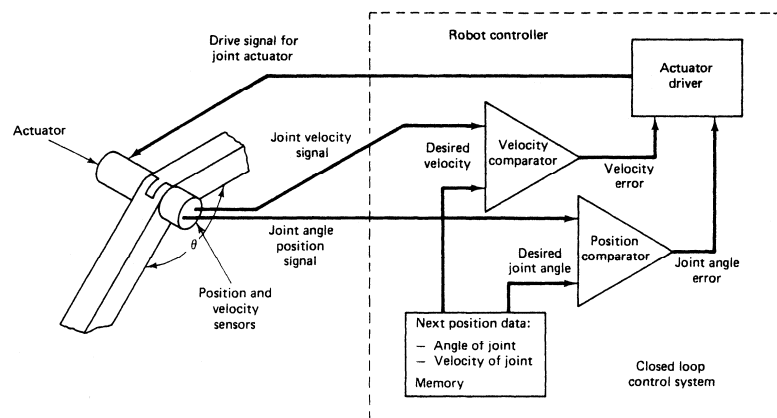
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Control Systems

• Open- and closed-loop systems:

- Robotic arms as a case study



Closed-Loop or Servo System

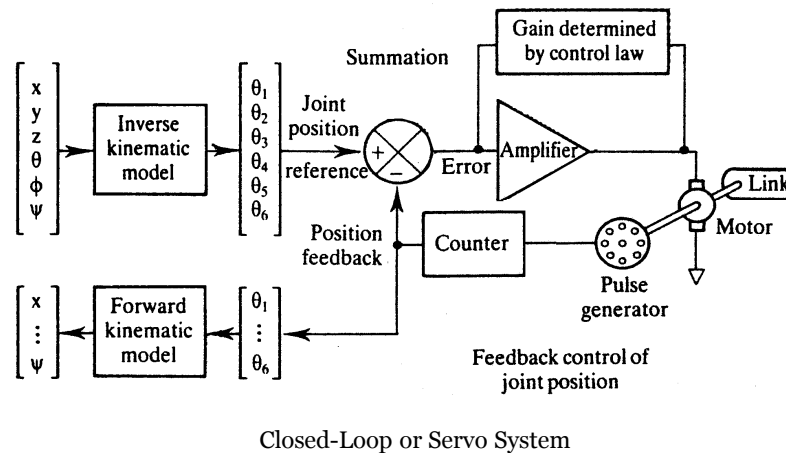
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Control Systems

• Open- and closed-loop systems:

- Robotic arms as a case study



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Control Systems

• Open- and closed-loop systems:

- Robotic arms as a case study

In the closed-loop control system the position of the robot arm is **continuously monitored** by a position sensor, and the power to the actuator is continuously altered so that the movement of the arm conforms to the desired path in both direction and velocity.

If a robot has 6 DOF or six axes, then the position of the joint in each degree of freedom is individually measured, and the electrical signals are sent by wire to the controller for analysis by the servo electronics. This sampling of joint positions and corresponding corrective action to the actuator occurs many times a second so that the visual effect is a smooth movement in the robot arm from one position to another.

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Control Systems

- **Open- and closed-loop systems:**

- Robotic arms as a case study

Advantages of closed loop or servo system:

- Flexible program control permits robots to be used in a wide variety of manufacturing jobs and extends the useful lifetime of the machine.
- Robots are capable of performing more complex manufacturing tasks
- Robots can execute multiple programs to handle varied manufacturing tasks.

Disadvantages:

- Bigger capital investment for a machine of this type is required.
- Maintenance staff must be highly skilled because of the increased technology in the work cell.

Control Systems

- **Open- and closed-loop systems:**

- Robotic arms as a case study

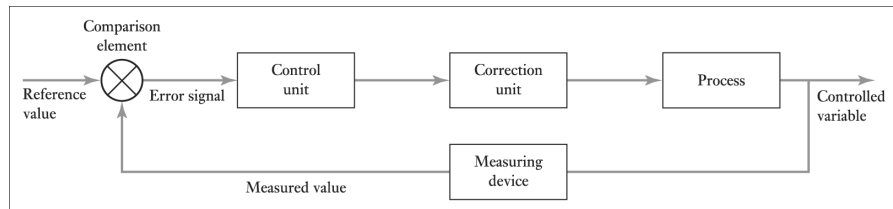
Applications:

The servo or closed-loop control system is used in any application in which path control is required, such as:

- Welding
- Coating
- Assembly operations

Control Systems

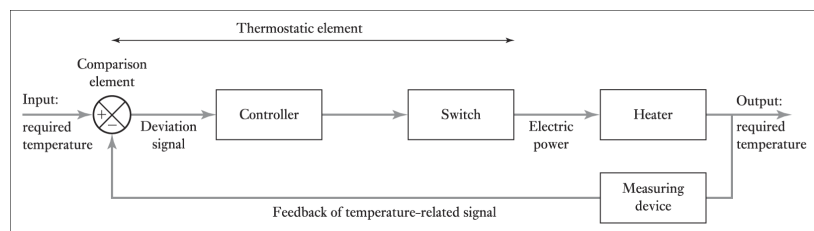
• Basic elements of a closed-loop system



The elements of a closed-loop control system

Control Systems

• Basic elements of a closed-loop system

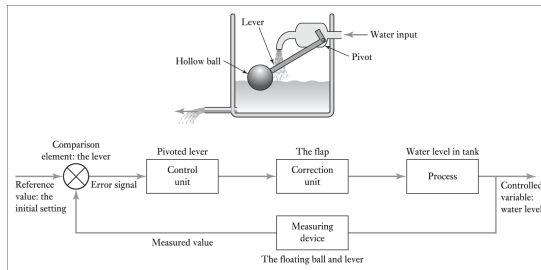


Heating a room: a closed-loop system

Controlled variable	The room temperature
Reference value	The required room temperature
Comparison element	Thermostatic element
Error signal	The difference between the measured and required temperatures
Control unit	Thermostatic element
Correction unit	The switch on the fire
Process	The heating by the fire
Measuring device	A thermometer

Control Systems

• Basic elements of a closed-loop system



The automatic control of water level

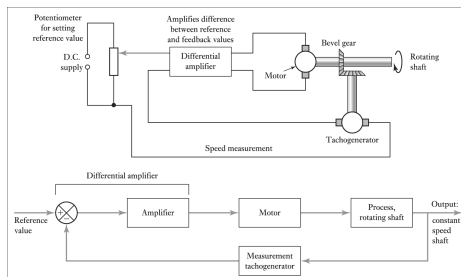
Controlled variable	Water level in the tank
Reference value	Initial setting of the float and lever position
Comparison element	The lever
Error signal	The difference between the actual and initial settings of the lever positions
Control unit	The pivoted lever
Correction unit	The flap opening or closing the water supply
Process	The water level in the tank
Measuring device	The floating ball and lever

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Control Systems

• Basic elements of a closed-loop system



Shaft speed control

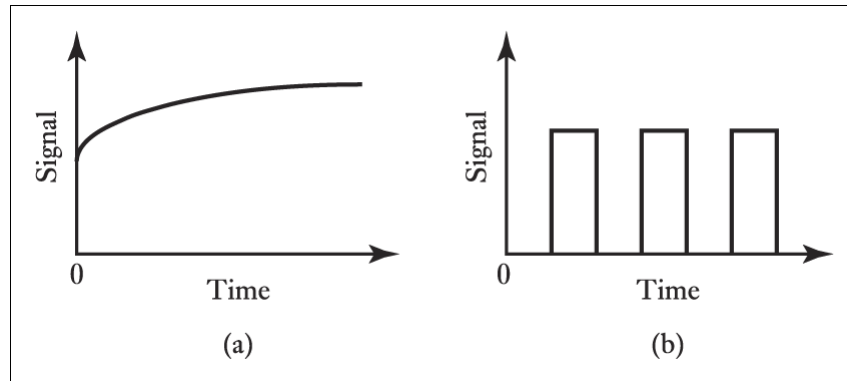
Controlled variable	Speed of rotation of shaft
Reference value	Setting of slider on potentiometer
Comparison element	Differential amplifier
Error signal	The difference between potentiometer and tachogenerator system outputs
Control unit	The differential amplifier
Correction unit	The motor
Process	The rotating shaft
Measuring device	The tachogenerator

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Control Systems

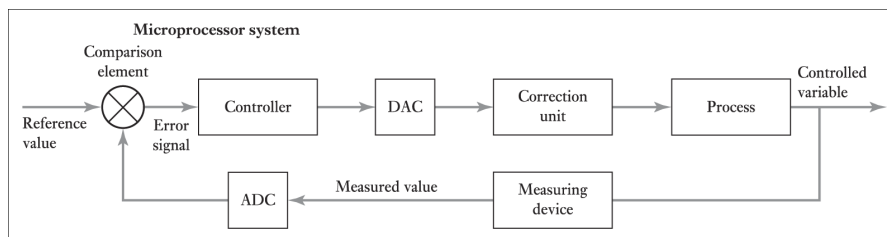
• Analogue and digital control systems



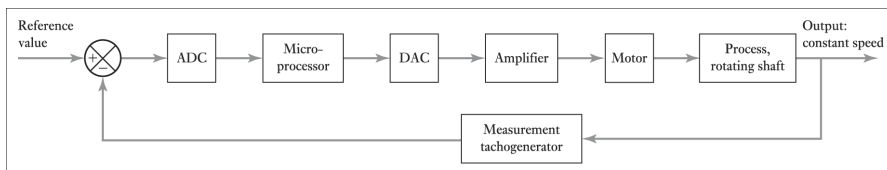
Signals: (a) analogue, (b) digital

Control Systems

• Analogue and digital control systems



The elements of a digital closed-loop control system



Shaft speed control

Control Systems

• Sequential controllers

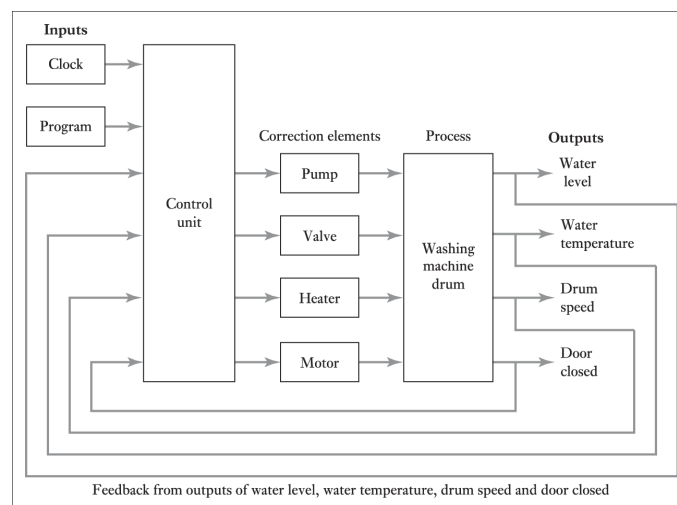
The term sequential control is used when control is such that actions are strictly ordered in a **time- or event-driven sequence**.

Such control could be obtained by an electric circuits with a set of **relays or cam-operated switches** which are wired up in such a way as to give the required sequence.

Such hard-wired circuits are now more likely to have been replaced by a **microprocessor-controlled** system, with the sequencing being controlled by means of a software program.

Control Systems

• Sequential controllers



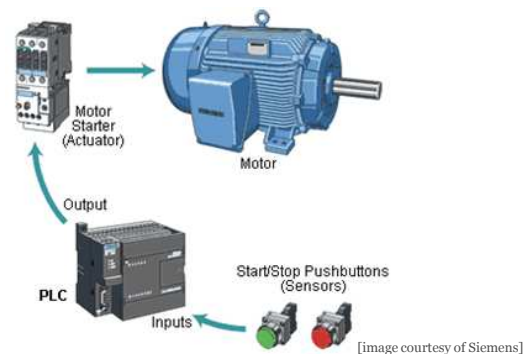
Washing machine system

Control Systems

- **Programmable Logic Controllers (PLCs)**

PLC is a microprocessor-based controller which uses programmable memory to store instructions and to implement functions such as logic, sequence, timing, counting and arithmetic to control events and can be readily reprogrammed for different tasks.

PLC is widely used in industry where on/off control is required.



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Outline

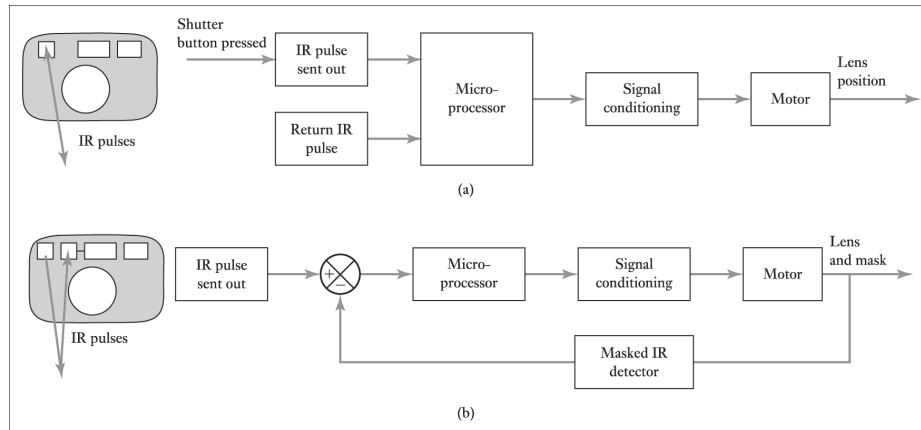
- What is mechatronics?
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- **Example of mechatronic systems**
- Summary

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Example of mechatronic systems

• The digital camera and autofocus



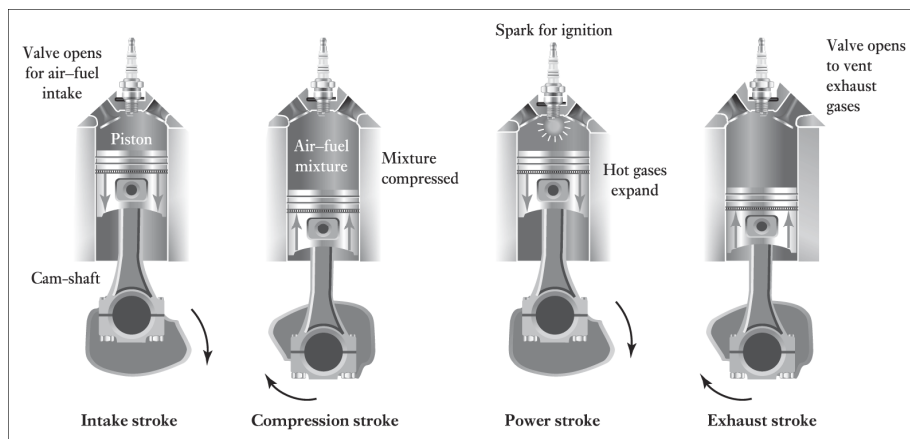
Autofocus: (a) open-loop system used in less expensive camera, (b) closed-loop system

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Example of mechatronic systems

• The engine management system



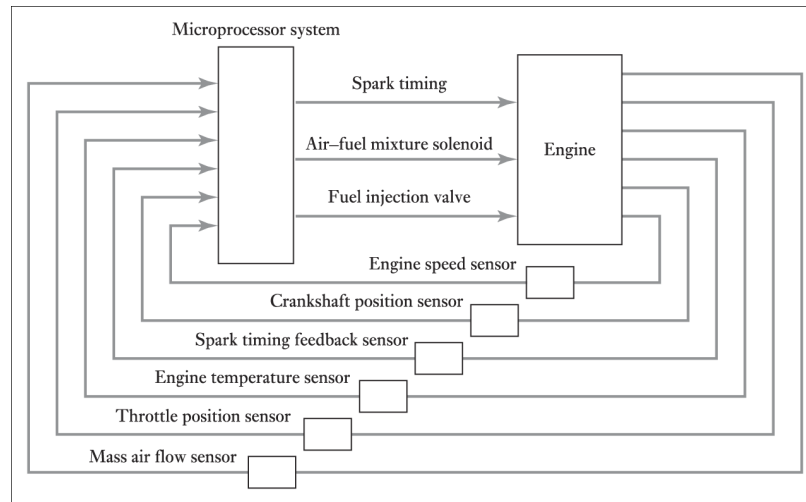
Four-stroke sequence

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Example of mechatronic systems

- **The engine management system**



Elements of an engine management system

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Outline

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Summary

- Mechatronics is a coordinated, and concurrently developed, integration of mechanical engineering with electronics and intelligent computer control in the design and manufacture of products and processes.
- Microprocessors are generally involved in mechatronics systems and these are embedded.
- Measurement systems are made up of three basic elements: sensor, signal conditioner and display.
- Control systems have two forms: open-loop and closed-loop.

Summary

Open-loop	Closed-loop
The output has no effect on the input	The output does have an effect on the input signal, modifying it to maintain an output signal at the desired value
Less accurate with non-deterministic behavior in case of perturbations	Relatively accurate in matching the actual to the required values
Relatively simple	More complex
Good reliability	Greater chance of breakdown as a consequence of the greater number of components
Lower cost	More costly

- **For reading:** Ch.1 Bolton's textbook.

Questions?